

Basic Mechanical Engineering (ME2001)

Date: May 31, 2025
Course Instructor(s)
Mohsin Yousuf

Final Exam

Total Time (Hrs): 3
Total Marks: 50
Total Questions: 5

Marks Distribution:

Q1	Q2	Q3	Q4	Q5
10	10	05	10	15

Roll No _____ Section _____ Student Signature _____

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1. Attempt all the questions, programmable calculators not allowed.
2. Show all the steps with the help of diagrams and answers with proper units.
3. In questions, hints are mentioned in *italic*.

CLO # 01: Calculate the moment of a force/couple.

Q1: The sum of the moments about point A due to the forces and couples acting on the bent bar as shown in Fig. 1 is zero. Answer the following: [10 marks]

- a) Determine the magnitude of the couple C using both scalar and vector formulation. [6]
- b) Determine the sum of the components about point B due to the forces and couples acting on the bar. Use the value of C as determined in part (a). [4]

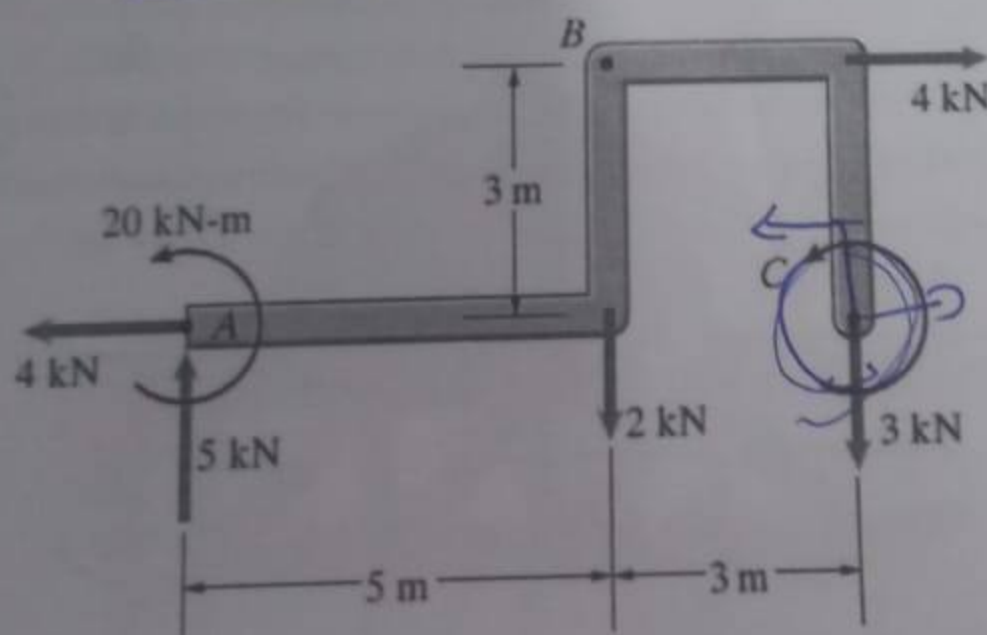


Figure 1. A Bent Bar

CLO # 02: Analyze static equilibrium analysis of a rigid body by applying Newton Laws of Motion and concept of dry friction.

Q2: The car shown in the Figure 1 below is of 1600 kg. The car has all-wheel drive and is just beginning to negotiate the 16° ramp. **Figure out** the minimum coefficient of static friction μ_s . Assume that slipping occurs at A and B simultaneously. [10 marks]

Remember to draw its FBD first.

The equations of friction would be using the same coefficient of static friction μ_s .



Figure 2. A Car about to drive the ramp at 16° at point A.

CLO # 03: Analyze absolute motion of rigid bodies in general plane motion.

Q3: The bar moves to the left with a constant velocity of v_A at end A as shown in the Fig. 3. **Figure out** the angular velocity ω of the bar as a function of its position x . [05 marks]

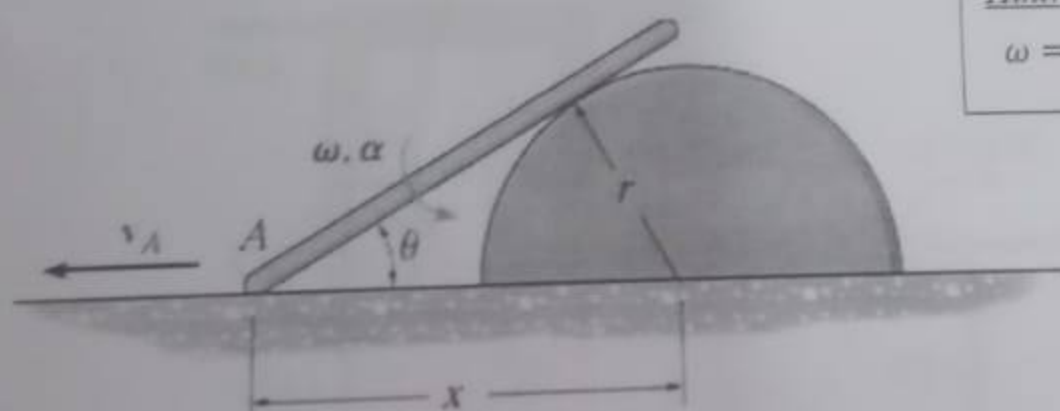


Figure 3. A Slender Bar moving to the left

Hint:

$$\omega = -\left(\frac{r}{x\sqrt{x^2 - r^2}}\right)v_A$$

CLO # 04: Analyze relative motion of rigid bodies in rotation and/or translation.

- Q4: Crank CB oscillates about C through a limited arc, causing crank OA to oscillate about O as shown in Fig. 4. When the linkage passes the position shown with CB horizontal and OA vertical, the angular velocity of CB is 2 rad/s counterclockwise. For this instant, figure out the angular velocities and angular acceleration of links AB and OA for this position. Solve by using vector algebra.

[10 marks]

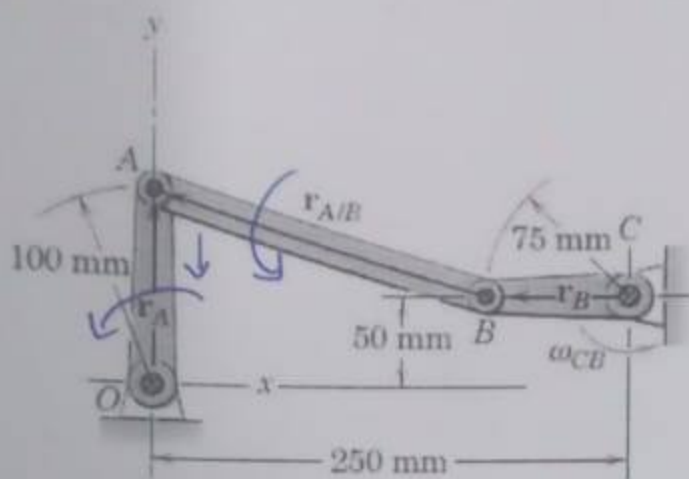


Figure 4. Ripple effect of Cranks pinned at C and O .

CLO # 05: Evaluate the internal forces in the members of a loaded truss and assess safety of the structure.

- Q5: The truss supports a 100 kN load at J as shown in the Fig. 5 below. The horizontal members are each 1 m in length.

[15 marks]

- Identify any zero-force member first by mentioning the rule.
- Use the method of joints to evaluate the axial forces in the members BC , CF , and FG .
- Verify them using method of sections.

There is pin at A and roller support at E .

All members have a circular cross-section having radius of 0.018 m . The ultimate tensile/compressive stress of the material used is 295 MPa .

Comment on the safety of the member i) BC and ii) the overall structure.

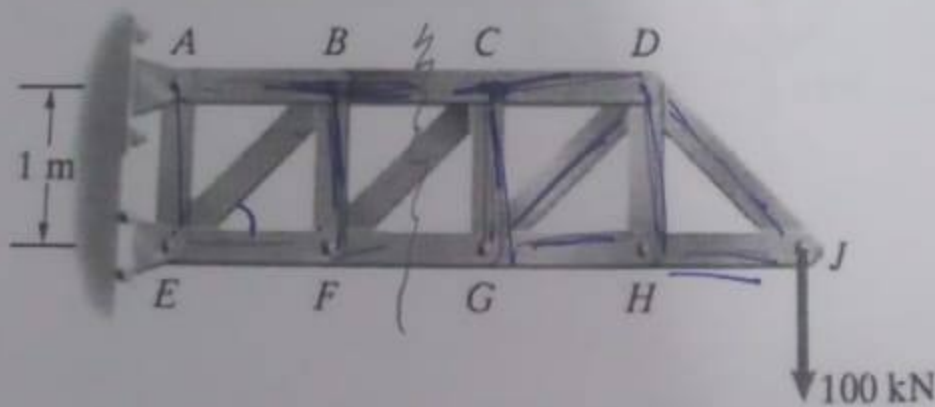


Figure 5. A Plane Truss

Basic Mechanical Engineering (ME2001)

Date: May 3, 2025

Course Instructor(s)

Mohsin Yousuf

Midterm-II Exam (Retake)

Total Time (Hrs):

Total Marks:

Total Questions:

1

40

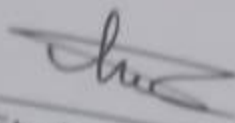
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23L-6026

Roll No

2B

Section


Student Signature

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1. Attempt all the questions, programmable calculators not allowed.
2. Attempt all parts of the same question together.
3. Show all the steps with the help of diagrams and answers with proper units.

CLO # 02: Analyze static equilibrium analysis of a rigid body by applying Newton Laws of Motion and concept of dry friction. [C4]

Q1: Determine the distance s to which the 90-kg painter can climb without causing the 4-m ladder to slip at its lower end A . The top of the 15-kg ladder has a small roller, and at the ground the coefficient of static friction is 0.25. The mass center of the painter is directly above her feet. [20]



Figure 1. A person climbing a ladder

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05: Evaluate the internal forces in the members of a loaded truss and assess safety of the structure. [C6]

The truss shown in the Figure 2 has pin at C and roller at K . **Determine** the force in the members DE and DF using the method of joints, when the force $P = 20 \text{ kN}$. **Verify** the axial force in DE using the method of sections.

[20 marks]

All members have a circular cross-section having radius of 0.02 m . **Evaluate** which material is suitable for the truss to avoid damage to the structure and compute its factor of safety as well.

Table 1. Ultimate Tensile Strength

Lead	Al	Iron
12 MPa	40 MPa	5 MPa

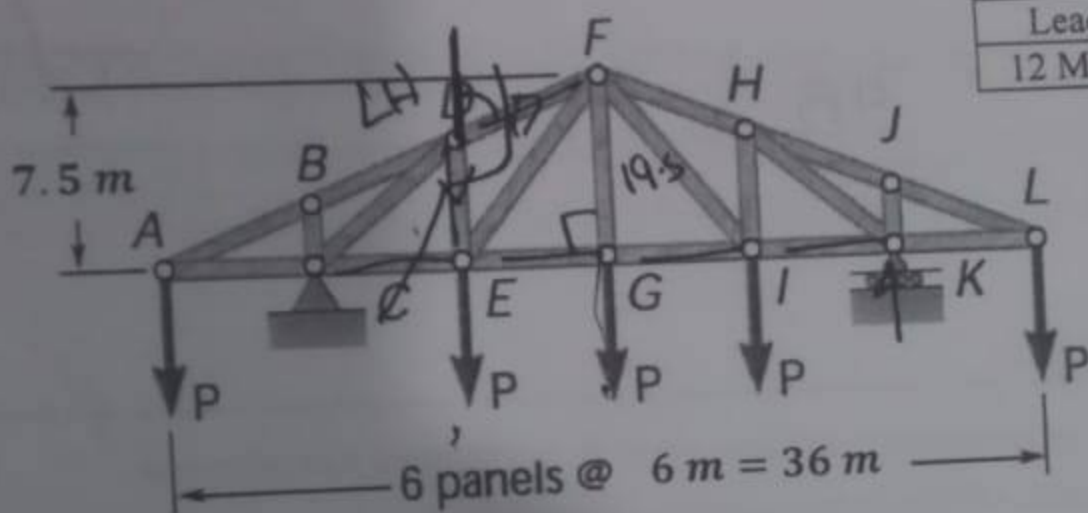


Figure 2. A truss with force P applied at 5 joints.